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flow of carriers.

In bipolar transistors, the flow is from emitter to collector through base, which exercises the control. There are two junctions involved, so carriers have to overcome two voltage barriers.

In MOS transistors, carriers flow from the source to drain without going through any voltage barrier. The voltage at the gate terminal controls the flow.

Due to the absence of voltage barriers, the heat dissipation in MOS is much lesser than in bipolar transistors. This allows millions of transistors to fit in a single die, which has considerable advantage over bipolar circuits.

MOS technology has become the predominant technology in digital integrated circuits. This is essentially because area occupancy, operating speed, energy efficiency and manufacturing costs have benefited and continue to benefit from the geometric downsizing that comes with every new generation of semiconductor manufacturing processes.

Most VLSI digital memory and microprocessor circuits are based on the MOS technology because:

1. Less chip area is demanded by an individual MOS transistor, which results in more functions in less area.
2. Critical defects per unit chip area are fewer for an MOS transistor because the MOS transistor requires fewer steps in fabrication.
3. MOS devices offer a high noise immunity.
4. Fewer transistors are required to realise a circuit function.
5. Fabrication cost is lower than bipolar one.

Q3. Where can I get IC PIC18F4431? How much does it cost?

Rupesh Chaudhary
Nepalgunj, Nepal



Sony Bravia LCD TV

A3. PIC18F4431 microcontroller is a product of Microchip Technology. You can buy it from them or their distributors RS Components or Element 14. For price details, visit the company's website at www.microchip.com

Q4. (a) What makes a maximum power-point tracking charge controller (MPPT) extract maximum power from solar cells?

(b) How do liquid-crystal display (LCD) and LED screens of televisions work? Why is an LED screen more energy efficient than LCD?

(c) Can energy be trapped from the radio waves and stored?

Santosh Kumar
Kakinada, Andhra Pradesh

A4. (a) Detailed information on maximum power-point tracking was published in EFY Dec. 2011 issue on page 20.

(b) The working and comparison between LCD and LED TVs were published in EFY June 2010 issue on page 18.

(c) Energy associated with information transmitted by radio waves is in microwatts at the receiving station. It can be extracted but so far the present technology has no provision of storing the energy.

Answers compiled by Col (Retd) N.C. Pande.

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